

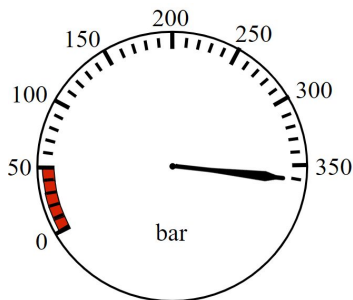
Name: _____



Emergency response worker

Task A: Uses of formulas

1) a) A firefighter enters the building with their gauge reading like so. At what bar should they turn around and start exiting?



$$t = \frac{1}{2}b + 30$$

b) A firefighter has an estimated turn-around pressure of 160 bar. How many bar of pressure did they have when they entered?

c) What is the difference in turn-around pressures between entering with 240 bar and entering with 300 bar?

$$t = \frac{1}{2}b + 30$$

d) Why do you think the formula includes + 30 instead of just being $t = \frac{1}{2}b$?

2) a) You are a police officer investigating an accident. Use the formula to calculate the speed of this car on a dry concrete surface ($f = 0.8$)

$$s = \sqrt{30df}$$



b) If a car were travelling 70 mph on a wet concrete surface ($f = 0.4$) how long would you expect the skid mark to be?

c) Lucas has used the formula incorrectly to calculate the speed of a car with a skid mark of 60 feet on a dry concrete surface ($f = 0.8$). Explain his error to him and correct his work.

$$s = \sqrt{30df}$$



My calculator tells me their speed was 262.9 mph



d) A police officer investigating an accident in a 20 mph zone with a dry concrete surface ($f = 0.8$) declares, "The length of the marks were 20 feet to the nearest 10 feet".

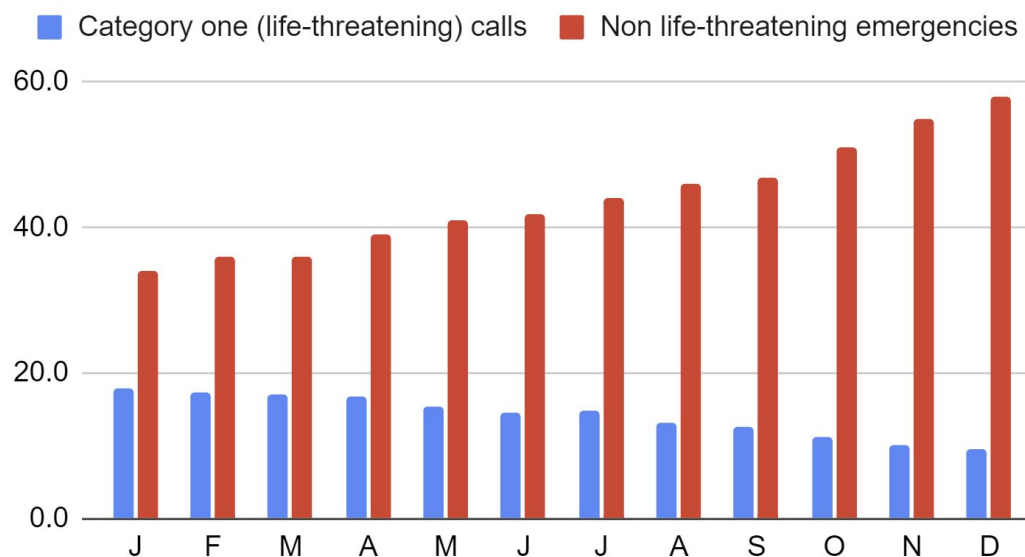
Explain why they cannot round like this when investigating this accident.

$$s = \sqrt{30df}$$



Task B: Uses of data

1) An ambulance station records their average response times to emergency calls one year. What patterns do you see in this data?



2) A regional fire service records the reasons for house fires over a year and produces this chart. What advice would you give this fire service? Write at least three sentences.

